Metals Market Index MMI

MMi Dashboard

Iron Ore Price Indices

Iron Ore Port Stock (FOT Qingdao)
IOPI62 62% Fe Fines RMB/t

822
-5 -0.60%
Dec 23rd, 2021

Iron Ore Port Stock (FOT Qingdao)
IOPI65 65% Fe Fines RMB/t

1015
-6.00 -0.59%
Dec 23rd, 2021

Iron Port Stock (FOT Qingdao)
IOPI58 58% Fe Fines RMB/t

602
-2 -0.33%
Dec 23rd, 2021

Iron Ore Seaborne (CFR Qingdao)
IOSI62 62% Fe Fines USD/dmt

124.00
-0.95 -0.76%
Dec 23rd, 2021

Iron Ore Seaborne (CFR Qingdao)
IOSI65 65% Fe Fines USD/dmt

134.15
-0.05 -0.04%
Dec 23rd, 2021

Iron Ore Port Stock (FOT Qingdao)
IOPLI 62.5% Fe Lump RMB/t

870
60 7.41%
Week Ending Dec 17th, 2021

Exchange Traded Contracts

DCE Iron Ore 62% Fines
I2201 (Jan) RMB/t (3pm close)

692.50
-1.00 -0.14%
Dec 23rd, 2021 (3pm close)

SGX Iron Ore (CFR Qingdao) 62% Fe Fines
December 21 USD/dmt

124.40
-1.35 -1.07%
Dec 23rd, 2021 (5.30 pm Print)

SHFE Rebar
RB2205 (Jan) RMB/t

4479
41 0.92%
Dec 23rd, 2021 (3pm close)

Freight Rates

C3, Tubarao - Qingdao USD/t

22.03
-0.17 -0.77%
Dec 22nd, 2021

C5, W. Australia - Qingdao USD/t

9.15
-0.21 -2.24%
Dec 22nd, 2021

Steel Price

Steel Rebar (China Domestic) RMB/t

4900
100 2.08%
Week Ending Dec 17th, 2021

Inventory Levels

Iron Ore Inventory at Chinese Ports (35)
million tonnes

152.72
2.15 1.43%
Week Ending Dec 17th, 2021

Steel Inventory in China
million tonnes

10.16
-0.69 -6.32%
Week Ending Dec 17th, 2021

Steel Price

Steel HRC (China Domestic) RMB/t

4940
140 2.92%
Week Ending Dec 17th, 2021

IRON ORE PORT STOCK INDEX (IOPI)

Dec 23rd, 2021		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne [1]						
Index	Fe Content	Price	Change	Change %	MTD	YTD	Low [2]	High [2]	Price	Change	Change %	MTD	YTD	Low [2]	High [2]
IOPI62	62% Fe Fines	822	-5	-0.6%	752	1146	587	1680	119.99	-0.66	-0.5%	109.39	166.50	84.25	247.30
IOPI58	58% Fe Fines	602	-2	-0.3%	551	920	454	1421	87.70	-0.23	-0.3%	79.81	134.29	64.78	210.83
IOPI65	65% Fe Fines	1015	-6	-0.6%	946	1349	843	1894	149.16	-0.78	-0.5%	138.67	196.77	122.83	278.61

IRON ORE SEABORNE INDEX (IOSI)

Dec 23rd, 2021		CFR Qingdao, USD/dry tonne						
Index	Fe Content	Price	Change	Change %	MTD	YTD	Low [2]	High [2]
IOSI62	62% Fe Fines	124.00	-0.95	-0.76%	112.77	162.84	93.75	230.50
IOSI65	65% Fe Fines	134.15	-0.05	-0.04%	125.09	187.86	108.20	262.95

MARKET COMMENTARY

DCE iron ore futures continued to drop today, and the main contract closed 692.5. an decrease only 0.65% throughout the day; the quotation in the morning was decreased by 5-10 yuan/mt over yesterday. some traders quotation is still stronger. steel mills purchased on demand. Today the overall market transaction atmosphere is acceptable. PBF at Shandong port dealt 790-800 yuan/mt, decreased 5-10 yuan/mt over yesterday；SSF at Shandong port dealt 490-495 yuan/mt, the same as yesterday；PBF at Tangshan port dealt 792-805 yuan/mt, decreased 5-10 yuan/mt over yesterday；PBF at Jiangnei port dealt 818 yuan/mt;

This week, SMM counted blast furnace operating rates at 77.35%, up 1.04% week-on-week and 0.44% month-on-month. Up to now, in addition to individual areas, the overall crude steel output does not exceed last year's target, Tangshan, Shandong, part of the blast furnace has resumed production, the recent blast furnace operating rate ushered in a trend to rise. Give mine price certain support, iron ore prices are still based on shock operation.

IRON ORE PORT LUMP INDEX (IOPLI)

Week Ending Dec 17th, 2021		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne [3]						
Index	Fe Content	Price	W-o-W	Change %	MTD	YTD	Low [2]	High [2]	Price	W-o-W	Change %	MTD	YTD	Low [2]	High [2]
IOPLI62	62.5% Fe Lump	870	60	7.4%	827	1373	730	1868	121.84	8.61	7.60%	115.56	190.66	101.00	262.00

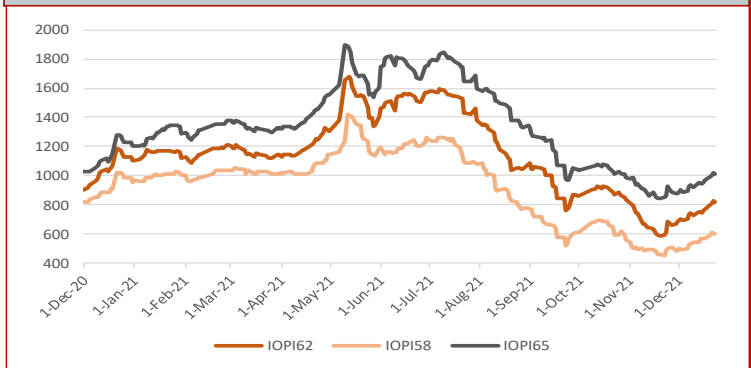
IRON ORE DOMESTIC CONCENTRATE SPOT PRICE ASSESSMENTS AND COMPOSITE INDEX

Week Ending Dec 17th, 2021				RMB/tonne (excluding tax) [3]				USD/tonne (excluding tax) [3]			
Province	Region	Product	Basis	This week	Change %	Low [2]	High [2]	This week	Change %	Low [2]	High [2]
Hebei	Hanxing	66% Fe Concentrate	Dry	895	4.1%	779	1645	140.61	4.09%	110.31	255.69
Hebei	Qian'an	65% Fe Concentrate	Dry	920	2.2%	780	1630	144.54	2.24%	110.51	251.57
Liaoning	Anshan	65% Fe Concentrate	Wet	740	2.8%	620	1310	116.26	2.80%	87.40	202.32
Shandong	Zibo	65% Fe Concentrate	Dry	910	4.6%	800	1752	142.97	4.62%	122.55	272.32

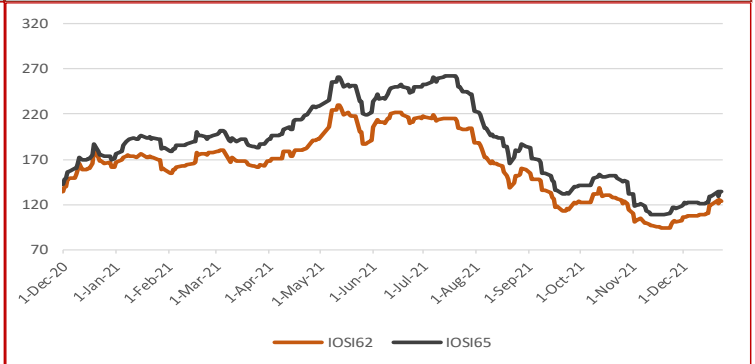
Week Ending Dec 17th, 2021	This week	Change %	Low [2]	High [2]
China Mines Concentrate Composite Index RMB/WT	819.20	4.38%	706.36	1511.22

[1] Exchange rate applied: RMB/USD = 6.4934 [2] Last 12 months
[3] Weekly exchange rate applied: RMB/USD 6.5044

IRON ORE PORT INDEX, FOT QINGDAO (RMB/WT)



IRON ORE SEABORNE INDEX, CFR QINGDAO (USD/DMT)



IRON ORE PORT STOCK INDEX MONTHLY, QUARTERLY AND YEAR-TO-DATE AVERAGES

Dec 23rd, 2021		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne						
Index	Fe Content	August	September	October	November	MTD	QTD	YTD	August	September	October	November	MTD	QTD	YTD
IOPI62	62% Fe Fines	1174	937	893	694	752	762	1146	170.18	135.90	135.31	100.58	109.39	110.41	166.50
IOPI58	58% Fe Fines	894	648	646	500	551	556	920	130.00	93.62	93.31	72.07	79.81	80.24	134.29
IOPI65	65% Fe Fines	1466	1152	1043	888	946	956	1349	213.59	214.41	214.06	129.88	138.67	139.56	196.77

IRON ORE SEABORNE INDEX MONTHLY, QUARTERLY AND YEAR-TO-DATE AVERAGES

Dec 23rd, 2021		CFR Qingdao, USD/dry tonne						
Index	Fe Content	August	September	October	November	MTD	QTD	YTD
IOSI62	62% Fe Fines	162.80	128.43	127.04	106.10	112.77	112.64	162.84
IOSI65	65% Fe Fines	192.73	148.09	147.37	121.55	125.09	128.89	187.86

FREIGHT RATES

Dec 22nd, 2021		FREIGHT RATES - DRY BULK US$/wet tonne				
Route	Designation		Change	Change %	Low [2]	High [2]
W. Australia - Qingdao	C5	9.15	-0.21	-2.24%	3.57	16.77
Tubarao - Qingdao	C3	22.03	-0.17	-0.77%	6.70	36.40

IRON ORE PORT LUMP INDEX MONTHLY, QUARTERLY AND YEAR-TO-DATE AVERAGES

Week Ending Dec 17th, 2021		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne [1]						
Index	Fe Content	August	September	October	November	MTD	QTD	YTD	August	September	October	November	MTD	QTD	YTD
IOPLI62	62.5% Fe Lump	1409	1065	999	800	827	908	1373	254.64	196.49	147.90	111.25	115.56	126.58	190.66

IRON ORE INDEX PREMIUMS/DISCOUNTS

Dec 23rd, 2021		PORT STOCK INDEX (RMB/WT)	
Index	Fe Content	Spread to IOPI62	% Spread to IOPI62
IOPI58	58% Fe Fines	-220	-26.76%
IOPI65	65% Fe Fines	193	23.48%

Dec 23rd, 2021		SEABORNE INDEX (USD/DMT)	
Index	Fe Content	Spread to IOSI62	% Spread to IOSI62
IOSI65	65% Fe Fines	10.15	8.19%

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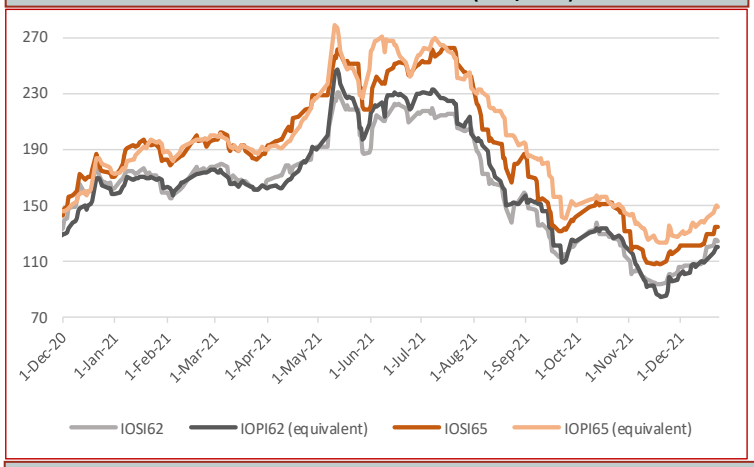
IRON ORE INDEX COMPARISONS (USD/DMT)



CHINA DOMESTIC COMPOSITE MINES INDEX (RMB/Wet Tonne, including VAT)

IRON ORE BRAND SPOT PRICE ASSESMENTS

Dec 23rd, 2021	PORT STOCK INDEX (RMB/WT)		
	Price	Change	Diff to IOPI62
Roy Hill	761	-5	-61
SIMEC Fines	693	-5	-129
PB Fines	790	-5	-32
Newman Fines	813	-5	-9
MAC Fines	760	-5	-62
Jimblebar Blended Fines	689	-5	-133
Carajas Fines	1001	-5	179
Brazilian SSF	785	-5	-37
Brazilian Blend Fines	843	-5	21
RTX Fines	705	-5	-117
West Pilbara Fines	733	-5	-89

Dec 23rd, 2021	SEABORNE INDEX (USD/DMT)		
	Price	Change	Diff to IOSI62
Roy Hill	119.50	-0.95	-4.50
SIMEC Fines	116.00	-0.95	-8.00
PB Fines	122.75	-0.95	-1.25
Newman Fines	123.90	-0.95	-0.10
MAC Fines	119.90	-0.95	-4.10
Jimblebar Blended Fines	113.25	-0.95	-10.75
Carajas Fines	153.55	-0.95	29.55
Brazilian SSF	127.75	-0.95	3.75
Brazilian Blend Fines	129.40	-0.95	5.40
RTX Fines	113.90	-0.95	-10.10
West Pilbara Fines	118.25	-0.95	-5.75

Dec 23rd, 2021	PORT STOCK INDEX (RMB/WT)		
	Price	Change	Diff to IOPI58
SSF	601	-2	-1
FMG Blended Fines	609	-2	7
Robe River	606	-2	4
Western Fines	612	-2	10
Atlas Fines	607	-2	5
Yandi	599	-1	-3

IRON ORE INDEX NORMALISATION DIFFERENTIALS

Port Stock Index Product Differentials (RMB/wet tonne)

	Applicable range	Value	Change
1% Fe	High Grade Fe 60 - 63%	10.00	-2.00
	High Grade Fe 63 - 64%	42.00	3.00
	High Grade Fe 64 - 65%	42.00	3.00
	High Grade Fe 65 - 65.5%	42.00	3.00
	Low Grade Fe	27.00	0.00
1% Alumina	High Fe Grade Al <2.25%	5.00	0.00
	High Fe Grade Al 2.25-4%	7.00	-2.00
	Low Fe Grade Al <2.25%	47.00	4.00
	Low Fe Grade Al 2.25-4%	43.00	0.00
1% Silica	High Fe Grade Si <4%	9.00	0.00
	High Fe Grade Si 4-6.5%	14.00	0.00
	Low Fe Grade	17.00	0.00
0.01% Phosphorus	High Fe Grade 0.09%<P<0.115%	14.00	-2.00
	High Fe Grade 0.115%<P<0.15%	14.00	-2.00
	Low Fe Grade 0.09<P<0.1%	5.00	0.00

Seaborne Index Product Differentials (USD/dry tonne)

	Applicable range	Value	Change
1% Fe	High Grade Fe 60 - 63%	3.50	0.00
	High Grade Fe 63 - 64%	0.50	0.00
	High Grade Fe 64 - 65%	0.50	0.00
	High Grade Fe 65 - 65.5%	0.50	0.00
1% Alumina	High Fe Grade Al <2.25%	0.25	0.00
	High Fe Grade Al 2.25-4%	4.25	0.00
1% Silica	High Fe Grade Si <4%	1.25	0.00
	High Fe Grade Si 4 - 6.5%	9.00	0.00
0.01% Phosphorus	High Fe Grade 0.09%<P<0.115%	0.25	0.00
	High Fe Grade 0.115%<P<0.15%	1.75	0.00

Port Stock Price Differentials to Qingdao Port for PB Fines (RMB/wet tonne)

Port	Value	Change	Port	Value	Change	Port	Value	Change	Port	Value	Change
Bayuquan	-100.00	0.00	Fangcheng	-25.00	0.00	Lanshan	0.00	0.00	Rizhao	0.00	0.00
Beilun	0.00	0.00	Jiangyin	-60.00	0.00	Lianyungang	0.00	0.00	Shekou	0.00	0.00
Caofeidian	0.00	0.00	Jingtang	0.00	0.00	Majishan	0.00	0.00	Taicang	-95.00	0.00
Dalian	0.00	0.00	Lanqiao	0.00	0.00	Qingdao	0.00	0.00	Tianjin	-40.00	0.00

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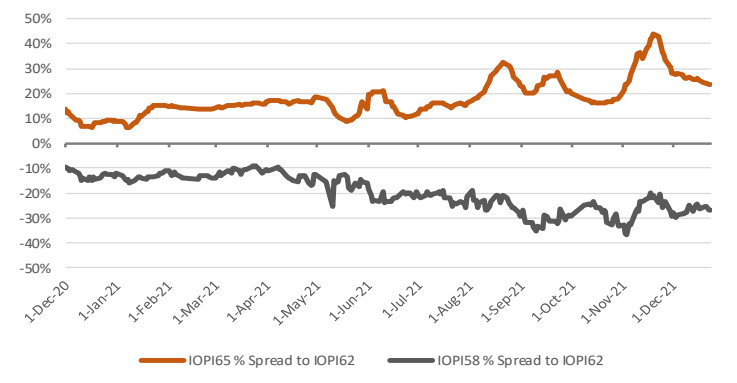
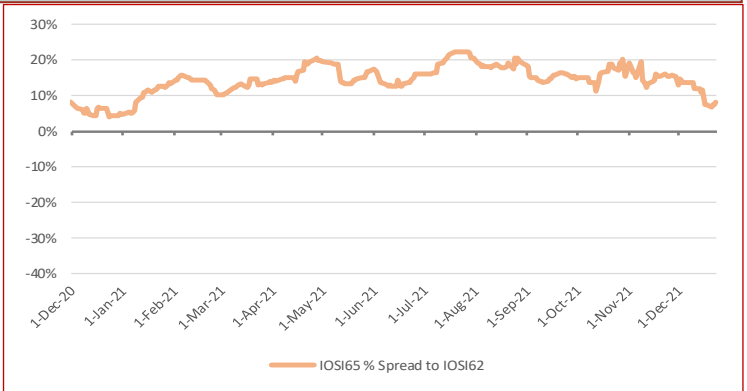
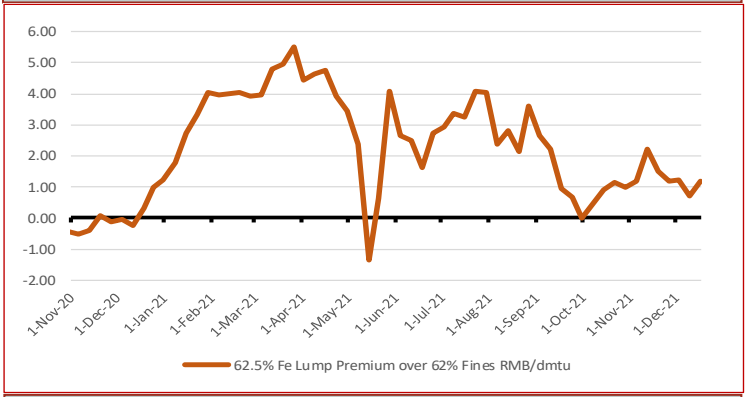
IRON ORE INDEX PREMIUMS/DISCOUNTS

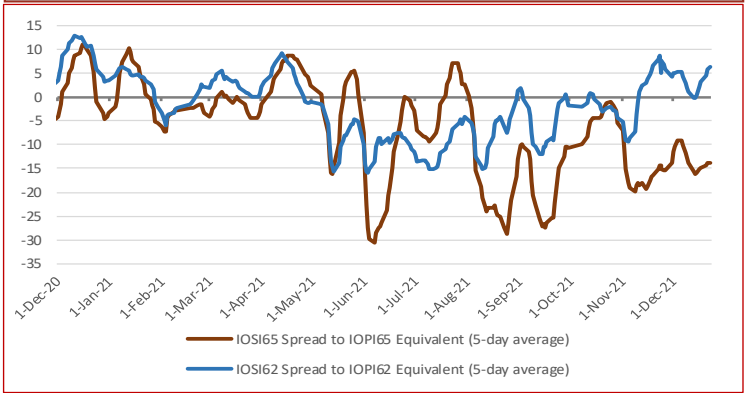




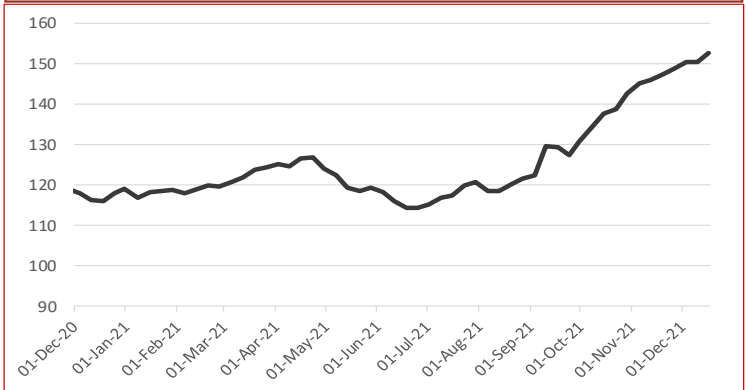
WEEKLY IRON ORE PORT STOCK LUMP PREMIUM (RMB/DMTU)



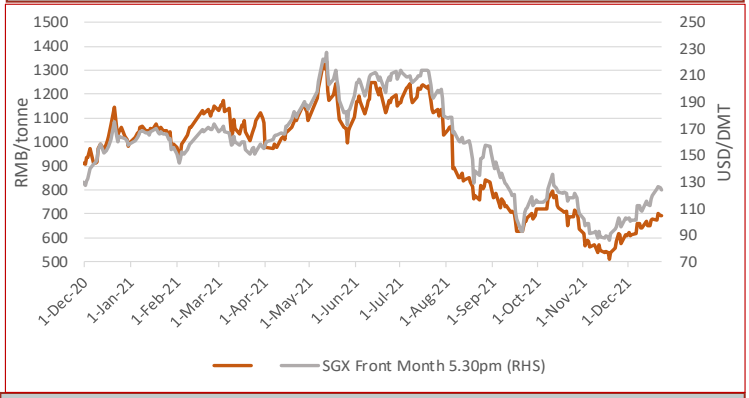
IRON ORE SEABORNE TO PORT STOCK PRICE SPREADS (USD/DMT)



TOTAL IRON ORE INVENTORIES AT CHINA PORTS (MILLION TONNES)



FUTURE TRADING—FRONT MONTH CLOSING PRICE



IRON ORE PORT INVENTORIES (MILLION TONNES)

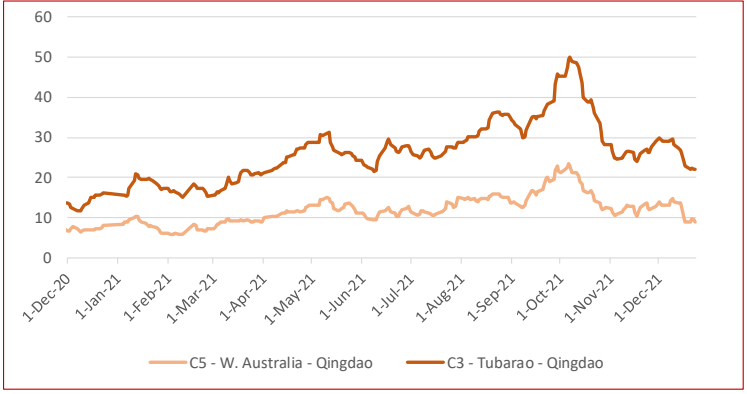
Week Ending Dec 17th, 2021

Province	This week	Change %	Low [2]	High [2]
Jingtang	15.54	1.24%	12.26	19.20
Qingdao	23.27	0.43%	9.41	23.96
Caofeidian	15.74	2.94%	11.25	16.25
Tianjin	11.18	-0.89%	7.14	11.38
Rizhao	17.21	5.71%	9.44	17.21
Total (35 Ports)	152.72	1.43%	98.80	152.72

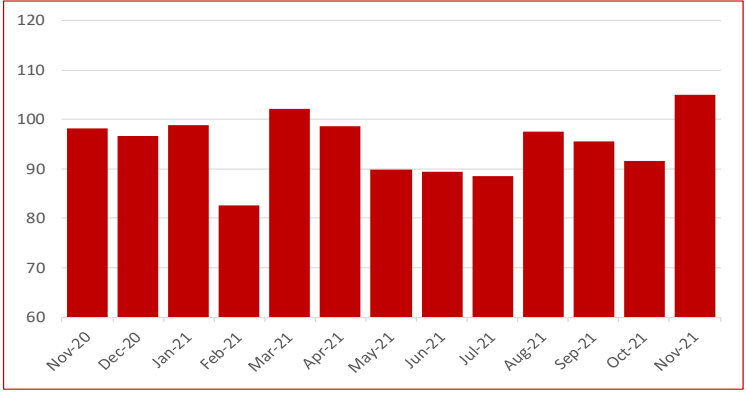
IRON ORE FUTURES CONTRACTS

	DCE (RMB/WMT)			SGX (USD/DMT)		
Closing Date	Dec 23rd, 3pm close			Dec 23rd, 5:30pm		
Contract	I2205	Change	Change %	Nov. 21	Change	Change %
Closing Price	692.50	-1.00	-0.14%	124.40	-1.35	-1.07%
Vol traded ('000 lots)	95.01	2.45	2.64%	2.81	0.02	0.56%
Open positions ('000 lots)	64.50	-0.69	-1.06%	21.38	-0.23	-1.08%
Day Low	676.0	-9.00	-1.31%	121.80	-4.17	-3.31%
Day High	696.0	-14.00	-1.97%	126.00	-2.70	-2.10%

DRY BULK FREIGHT RATES (USD/MT)



TOTAL CHINA IRON ORE IMPORT VOLUMES (MILLION TONNES)



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STEEL SPOT MARKET PRICES—CHINA

Steel Spot Market RMB/tonne

Product	2021/12/17	Change	Change %
ReBar HRB400 ϕ18mm	4,900	100	2.08%
Wirerod Q300 ϕ6.5mm	5,140	80	1.58%
HRC Q235/SS400 5.5mm*1500*C	4,940	140	2.92%
CRC SPCC/ST12 1.0mm*1250*2500	5,610	20	0.36%
Medium & Heavy Plate Q235B 20mm	5,100	0	0.00%
GI ST02Z 1.0mm*1000*C	5,660	10	0.18%
Colour Coated Plate	8,200	-50	-0.61%

CHINESE STEEL MILL PROFITABILITY

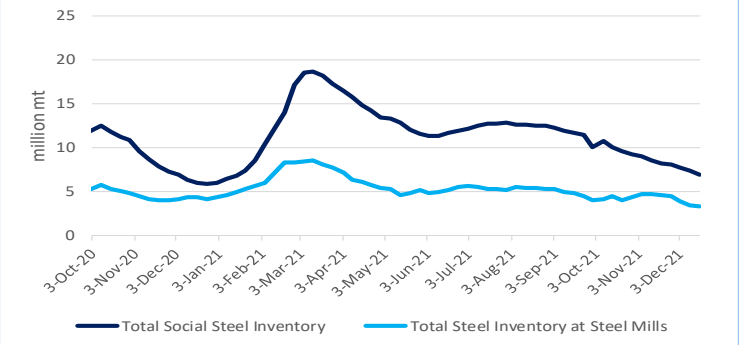
SMM Tracking of Steel Mill P&L - Rebar and Hot-rolled Coil (RMB/tonne)

Category	Price	Change (WoW)	Note
MMi (Fe 62%), USD/mt exluding tax	120.05	13	Mmi CFR Equivalent index for 1st Feb
Coke	2,900	290	2nd grade met coke, Tangshan, incl. tax
Steel Scrap	3,440	200	steel scrap (6mm) in Zhangjiagang, exl. tax
Billet Cost	3,765	107	Q234, incl. tax
Rebar cost - Blast furnace	4,390	121	calculated based on theoretical weight, incl. tax
Rebar profit - Blast furnace	450	-11	based on Shanghai prices, incl. tax
Hot-rolled coil cost - Blast furnace	4,571	122	based on actual weight, incl. tax
Hot-rolled coil proft - Blast furnace	349	-12	based on Shanghai prices, incl. tax

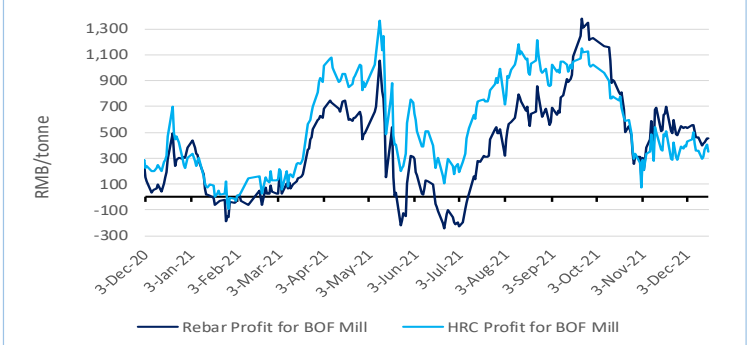
Note: 1. Costs in the table are caluclated based on todays market prices and facout our management, sales, financial and depreciations fe

2. The cost refers to average cost in the industry based on SMM's survey of small, medium and large mills in China

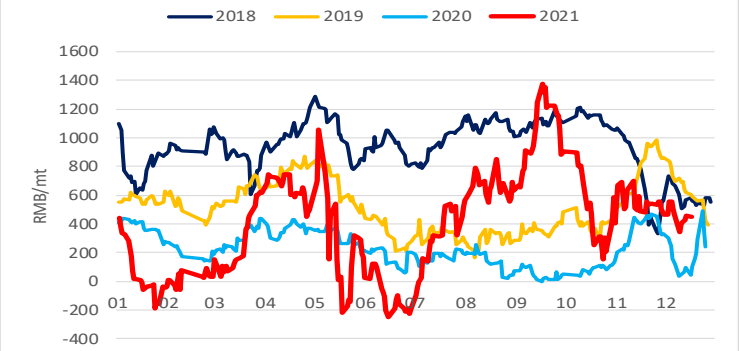
CHINESE STEEL INVENTORIES



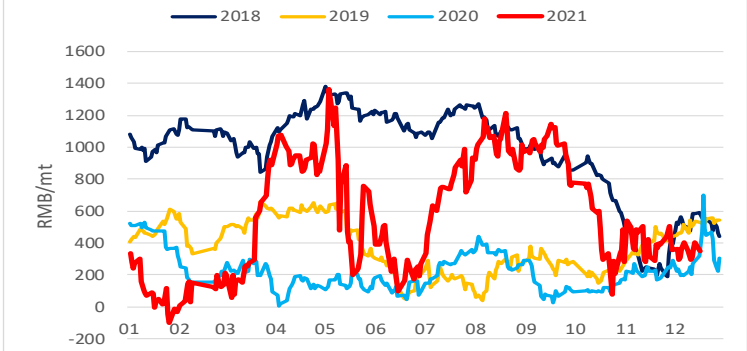
CHINESE STEEL MILL PROFITABILITY



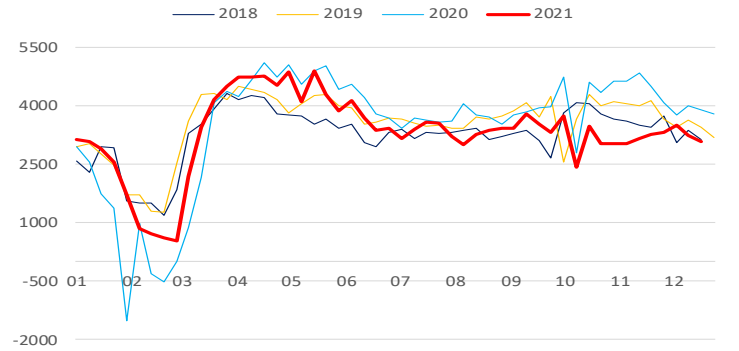
CHINESE STEEL MILL PROFITABILITY—Rebar



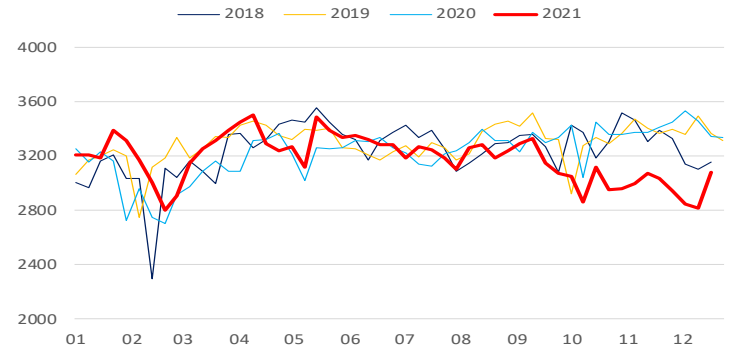
CHINESE STEEL MILL PROFITABILITY—Hot-rolled Coil



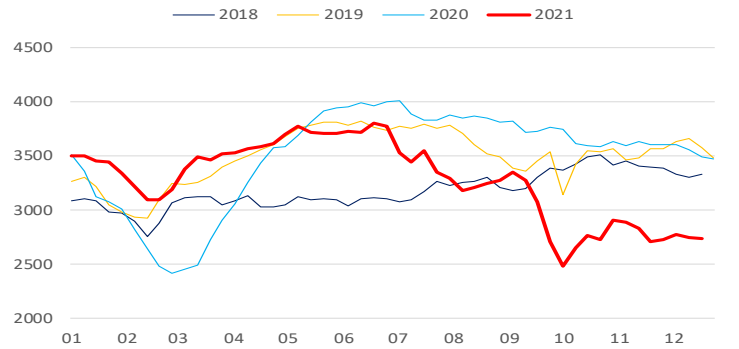
CHINESE STEEL CONSUMPTION—Rebar



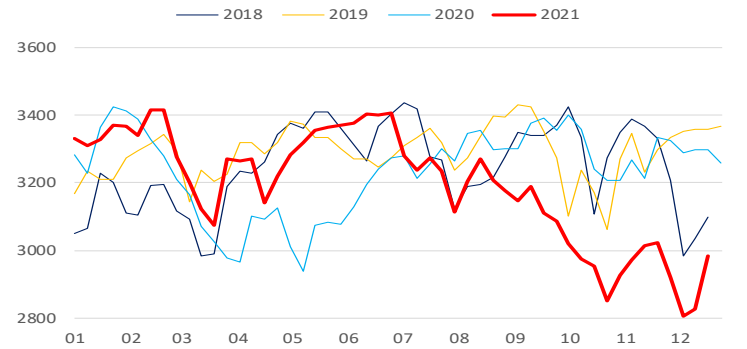
CHINESE STEEL CONSUMPTION—Hot-rolled Coil



CHINESE STEEL PRODUCTION—Rebar



CHINESE STEEL PRODUCTION—Hot-rolled Coil



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IRON ORE INDEX SPECIFICATIONS, COMPILATION RATIONALE AND DATA EXCLUSIONS

Iron Ore Index Specifications (Port and Seaborne)

	65% Fe Fines	62% Fe Fines	58% Fe Fines	62.5% Fe Lump
Fe %	65.00	62.00	58.00	62.50
Alumina %	1.40	2.25	2.25	1.50
Silica%	1.50	4.00	5.50	3.50
Phosphorus %	0.06	0.09	0.05	0.08
Sulphur %	0.01	0.02	0.02	0.02
Moisture %	8.00	8.00	9.00	4.00
Sizing	Granular size below 10mm for at least 90% of cargo; maximum of 40% below 150 micron			Size below 6.3mm max 15% Size above 31.15mm max 25%
Pricing Point	Qingdao Port (FOT and CFR respectively)			FOT Qingdao Port
Timing (Seaborne)	Loading within 4 weeks, Delivery within 8 weeks			Delivery within 2 weeks
Payment Terms	L/C at sight			L/C at sight or CAD

Iron Ore Index Compilation Rationale and Data Exclusions

MMi iron ore indices are compiled from data provided by companies that are part of the iron ore supply chain and involved in spot market transactions. The indices are calculated using detailed transaction-level data submitted to MMi by these companies. This data is normalised to the appropriate specifications and screened to remove outliers before volume-weighted average prices are calculated from the remaining core set of data.

For more details on MMi's iron ore methodology please download the guide published on our website at: www.mmiprices.com

Data Exclusions*

	62%	58%	65%
Port Index	0	0	0
Seaborne index	0		0
Lump Index 62.5	0		

* Number of price submissions for iron ore indices that were excluded from index calculations today as they were anomalous and could not be verified

IRON ORE DOMESTIC CONCENTRATE INDEX CALCULATION METHODOLOGY

The compilation method for price index generally refers to the compilation method of CPI price index and other price indices, breakdown the price data and calculate the average value according to a certain method, taking the vertical axis as the regional composite index (average of different grade index) and the horizontal axis as the grade composite index (average of different regional index), a total composite index for domestic ore can be output ultimately. The process system is also adopted in the calculation i.e. each sub-index can be obtained as well.

AVERAGE IRON ORE SPECIFICATIONS APPLIED FOR BRAND PRICE ASSESSMENTS

PORT STOCK BRANDS

Dec 23rd, 2021	Specifications applied for 62% brand assessments				
	Fe	Alumina	Silica	Phos	Moisture
Roy Hill	60.88%	2.35%	4.45%	0.055%	9.26%
SIMEC Fines	60.00%	2.30%	6.30%	0.060%	6.00%
PB Fines	61.64%	2.31%	3.87%	0.099%	9.45%
Newman Fines	62.49%	2.31%	3.98%	0.090%	7.72%
MAC Fines	60.43%	2.38%	4.64%	0.078%	7.57%
Jimblebar Blended Fines	60.87%	2.95%	4.53%	0.111%	7.32%
Carajas Fines	64.90%	1.39%	1.82%	0.080%	7.78%
Brazilian SSF	62.00%	1.00%	6.50%	0.040%	6.00%
Brazilian Blend Fines	62.56%	1.58%	4.79%	0.091%	8.72%
RTX Fines	61.00%	3.10%	4.50%	0.135%	7.50%
West Pilbara Fines	60.10%	2.30%	4.70%	0.075%	8.50%

Dec 23rd, 2021	Specifications applied for 58% brand assessments				
	Fe	Alumina	Silica	Phos	Moisture
SSF	56.49%	3.20%	6.19%	0.065%	9.18%
FMG Blended Fines	58.20%	2.43%	5.54%	0.057%	8.29%
Robe River	56.44%	3.16%	5.73%	0.042%	8.44%
Western Fines	57.88%	2.87%	7.50%	0.062%	7.45%
Atlas Fines	58.00%	1.85%	5.50%	0.090%	9.00%
Yandi	56.87%	1.58%	6.41%	0.042%	9.53%

SEABORNE BRANDS

	Specifications applied for 62% brand assessments				
	Fe	Alumina	Silica	Phos	Moisture
Roy Hill	60.70%	2.30%	4.90%	0.055%	8.00%
SIMEC Fines	60.00%	2.30%	6.30%	0.060%	6.00%
PB Fines 62%	62.00%	2.60%	4.30%	0.090%	9.50%
Newman Fines	62.30%	2.40%	4.40%	0.090%	8.00%
MAC Fines	60.80%	2.35%	4.70%	0.080%	7.80%
Jimblebar Blended Fines	60.50%	3.00%	4.50%	0.120%	8.00%
Carajas Fines	65.10%	1.50%	1.70%	0.080%	8.50%
Brazilian SSF	62.00%	1.00%	6.50%	0.040%	6.00%
Brazilian Blend Fines	63.00%	1.50%	5.00%	0.070%	7.00%
RTX Fines	61.00%	3.10%	4.50%	0.135%	7.50%
West Pilbara Fines	60.10%	2.30%	4.70%	0.075%	8.50%

BLOOMBERG TICKERS

PORT STOCK INDICES

	FOT Qingdao (RMB/wet tonne)	CFR Qingdao Equivalent (USD/dry tonne)
IOPI62	IRCNQ001	IRCNQ004
IOPI58	IRCNQ002	IRCNQ005
IOPI65	IRCNQ003	IRCNQ006
IOPLI62	IRCN0036	IRCN0037

SEABORNE INDICES

	CFR Qingdao (USD/DMT)
IOSI62	IRCN0034
IOSI65	IRCN0035

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